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Invited Review

Optimization in multimodal freight transportation problems: A Survey

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ABSTRACT

Since the rise of globalization, international trade has grown constantly and the demand of transportation services grew with it. Multimodal transportation is a natural evolution of the classical unimodal road transportation, and is a mandatory choice for intercontinental shipments. Optimization techniques always found a challenging but stimulating ground for applications in transportation, and the increase in the number of commodities that are transported every year all around the globe enhanced the interest and the usefulness of operational research methodologies, which are needed to properly manage complex transportation systems. The literature on multimodal transportation optimization is flourishing with interesting and multifaceted problems. Meanwhile, new emerging technologies generate new challenges for researchers. The objective of this literature review is to provide a picture of the state of the art on multimodal freight transportation optimization, for the different combinations of modes. An overview of recently published studies is given, identifying currently ongoing topics and emerging trends for future research.

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1. Introduction

Freight transportation is a key aspect of supply chain management, especially in an increasingly globalized economy. The increase in quantities and distance of international trades, and therefore of transportation of goods, made the typical unimodal road transportation a sub-optimal solution when not an infeasible one (intercontinental transportation requires at least another modality). We will use the two main definitions of transportation with multiple modes as in SteadieSeifi, Dellaert, Nuijten, Van Woensel, & Raoufi (2014). *Multimodal* freight transportation is often defined as the transportation of goods carried out by a sequence of at least two different modalities. Goods can be transported in various transportation units (crates, pallets, cages, containers) or as loose packages loaded directly on transportation means, such as vehicles, vessels, aircrafts. This is the broader and more general definition since it highlights the main features of transporting goods with multiple modalities. A more specific, often used, definition is *intermodality*. Intermodal freight transportation is defined as the transportation of goods by a sequence of at least two different modalities without changing the load unit during the transport chain. In intermodal transportation the goods are loaded at the origin place (usually the shipper warehouse) inside the load unit (the most common example is the container). The transportation unit is then moved between modalities until it reaches the final delivery location, where goods are unloaded. Intermodality exploits economies of scale and hastens transshipment operations, but requires proper facilities and equipment for the handling of transportation units. Intermodal return logistics is an issue to be tackled: due to traffic and trade imbalances, some countries are mainly exporters while others are natural importers and, thus, the need rises for empty transport units to be moved from import areas back to export areas. Fig. 1 shows the difference between multimodality and intermodality.

While the definitions remark particular aspects of the transportation, the main focus is the use of more than one modality in order to move freight from one location to another. In this work, when talking about transportation with multiple modalities, the term *multimodal* will be used since it represents the wider and more generic definition. Any specific definition will be used when needed, in order to better understand the setting of the study.

Multimodal transportation is divided in three phases

- *Pre-haul* is the transportation from the starting location to an interchange terminal where the shipment switches modality; it is usually carried out by road transportation.
- *Long-haul* is the leg which covers the longer distance. Long-haul leg transport is carried out by means of rail, maritime, inland waterways or air transportation. It is usually less expensive

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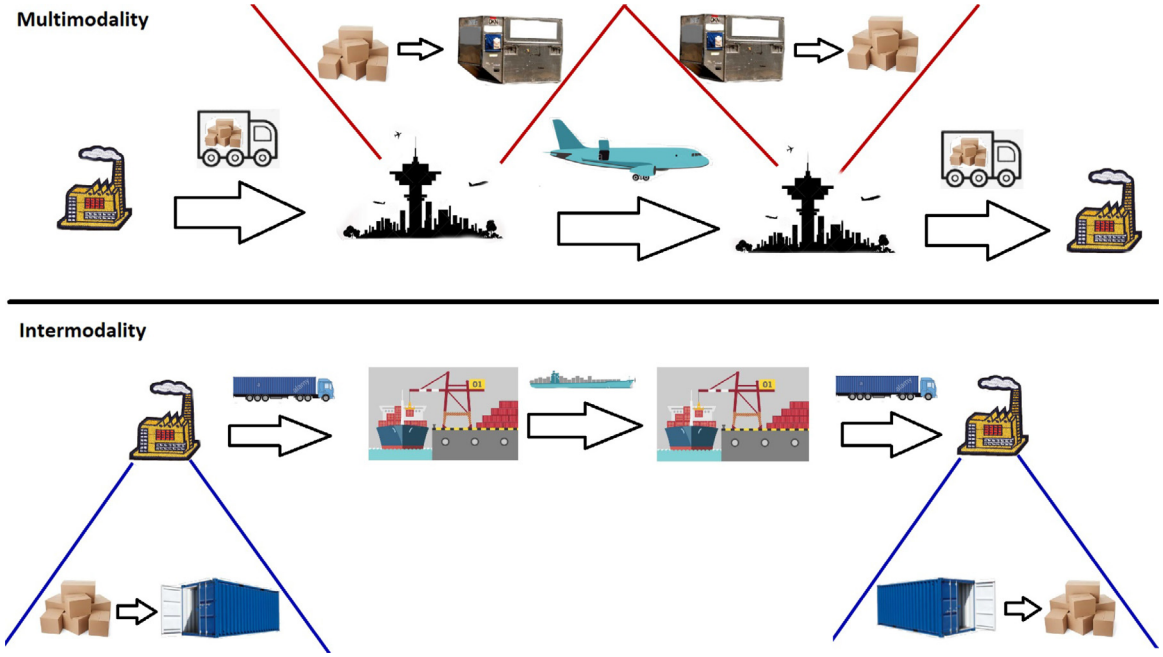


Fig. 1. Multimodal versus intermodal transportation.

when compared to road transportation for long distances and is often mandatory.
- *End-haul* is the so-called *last mile*, i.e. the last leg to the delivery location and, as in pre-haul, is carried out by road transportation in most of the cases.

In long-haul we can use more than one modality. For instance, if we consider an intercontinental transportation from China to the U.S., what usually happens is that pick-up transport is carried out by road from the origin to the nearest multimodal exchange terminal where containers are loaded to trains directed to one port. There, the shipment is moved through a maritime service and once arrived in the U.S., depending on the delivery location, it can again be moved through a rail transportation service to a multimodal terminal from where it is delivered to the final destination by road. Policy makers, like the European Commission in the White Paper on Transport 2010, pointed out the importance of creating a transportation network capable of fostering the development of multimodality in order to decrease emissions and congestion, improving efficiency and redistributing freight transportation between modalities. Modal split, however, is still largely unbalanced in favour of road transportation. In fact, in the last 30 years very little steps have been made in reducing the percentage of goods transported by road. According to EUROSTAT, in 2018, across the 28 countries members of the EU, 76.50% of freight was transported by road, 18.00% by rail and the remaining 5.50% by inland waterways.

The literature on multimodal transportation is very wide and in constant growth. The goal of this paper is to provide a review of the literature on optimization models for multimodal transportation. In particular, we aim at exploring the different problems and challenges that characterize each combination of multimodal transportation. We focus on classifying the literature based on the modality used in the long-haul leg. There are two main reasons for this choice. First, an optimization problem defined for a certain combination of modalities usually presents specific aspects that are typical of the modalities involved. A classification of the studies based on the modality used in the long-haul leg should help interested researchers to easily identify the related literature. Second, while some decision problems are common to different combinations of modalities, others are more related to certain modalities, as witnessed by the information reported in Table 1. The table is organized as follows. Decision problems are classified as operational, tactical and strategic. Operational problems are often easy to identify as they deal with daily or routine operations. Strategic problems often focus on planning investment decisions, mostly related to the infrastructures. Tactical problems tackle the optimal use of the already existing infrastructure by choosing services and associated transportation modes, deciding their capacities, and planning their itineraries and frequencies. Some problems can have a tactical or strategic nature, depending mostly on the time scale and horizon considered. The boundary between them is not always easy to draw and not every study is clearly categorized as tactical or strategic. Thus, we decided to classify the papers for their contribution to operational and tactical/strategic problems. In each class, major decisions are identified through those tackled in the papers reviewed in this literature review. The number of papers dealing with each decision per modality is provided in Table 1.

In Table 1 decisions are classified as follows:

- LOCATION: Papers studying problems with the main and sole focus on optimizing the location of facilities are included in this class. Problems where the location is part of a general network definition are classified as Network Design problems.
- NETWORK DESIGN: Papers that optimize the definition of networks are included in this class. Mainly Network and Service Network Design decisions are hereby included.
- SCHEDULING: Papers in this class study problems on service schedule, transshipment schedule, yard operations schedule, shipment and transport schedule.
- TRANSPORTATION: A paper is included in this class when the main decision is to define the best set of services used to transport commodities from one location to another. Sometimes (e.g. Xiong & Wang (2014)) these problems are called "route selection" or "route planning" problems. We found this kind of decision in both operational and tactical/strategic problems.
- FLEET RENEWAL: Papers where the renewal of a fleet is the main focus are included in this class.
- RESOURCE ALLOCATION: These problems deal with the decision about how resources should be allocated to different locations/operations.

Table 1
Classification of papers based on decision and modality.

	Decision	Air	Rail	Maritime	Multi
Tactical&Strategic	LOCATION	3	7	0	1
	NETWORK DESIGN	2	9	12	17
	SCHEDULING	0	0	3	4
	TRANSPORTATION	0	2	0	0
	FLEET RENEWAL	0	0	1	0
	RESOURCE ALLOCATION	0	0	0	1
Operational	TRANSPORTATION	2	4	3	6
	VESSEL SCHEDULE RECOVERY	0	0	6	0
	ROUTING (multiple)	5	3	1	3
	ROUTING (single)	11	0	0	0

- VESSEL SCHEDULE RECOVERY: This is one of the most critical problems faced in maritime transportation. Papers dealing with the recovery of the vessel schedule after disruptive events are included in this class.
- ROUTING: The main decision is to identify the route taken by either a single vehicle or a set of multiple vehicles. Whenever a paper aims at optimizing the route of one or more vehicles it is included in this class.

For each modality used in the long-haul leg we analyze the main decisions modeled in the analyzed papers, the nature of the problems tackled (operational, tactical and strategic) and point out possible future fields of research. In addition, for each modality, we provide the mathematical formulation of the operational and/or tactical/strategic problem that is most commonly studied in the literature, to give interested readers a starting point for further studies. For the formulations presented, we did not follow the notation used by the authors in the original works, but instead adopted a common notation with the objective of highlighting the common characteristics among the problems.

The paper is organized as follow. Section 2 presents some general information about the literature review. In Section 3, 4 and 5 we explore the papers where the long-haul transport is carried out on air, rail and sea, respectively, while 6 considers the use of more than two modalities.

Table 2
Frequency of keywords.

Keyword	Frequency
Transportation	71
Intermodality	42
Network	38
Drone	27
Routing	22
Logistics	15
Container	12
Emissions	12
Synchromodality	11
Multicommodity	11
Multimodality	10

Table 3
Frequency of keywords (2015–2020).

Keyword	Frequency
Transportation	39
Drone	27
Intermodality	24
Network	21
Emissions	12
Multicommodity	11
Synchromodality	11

2. Overview

The review covers 118 papers, most of which published in international journals, with a particular focus on papers published from 2014. In fact, an extensive review of the literature preceding 2014 can be found in SteadieSeifi et al. (2014). Papers published before 2014 that are considered to be particularly worth mentioning, either due to the number of citations or the relevance to the topic, are included in this literature review. Fig. 2 shows the distribution of papers by year of publication. The majority of papers was published after 2014, with a high concentration between 2015 and 2017.

The search of papers was made starting from the following keywords: intermodal transportation optimisation, multimodal transportation optimisation, intermodal network optimisation, multimodal network optimisation, synchromodal transportation optimisation, synchromodal network optimisation. From the first results, a second search considered a wider range of keywords, including problem-specific words (location, hub location, network design, service network design, multimodal commodity and multicommodity flow, supply chain optimisation) or modality-specific words (maritime, rail-road, truck-drone, truck-inland waterways). Then, starting from papers which were considered as interesting, the search was expanded either by searching the keyword used in the identified set of papers, or by exploring the related literature. With few exceptions, papers not addressing problems involving multiple modalities were discarded. The frequency with which the major keywords appear is reported in Table 2. It is to be noted that keywords are processed in order to be more easily categorized together (e.g the keywords "multimodal", "multimodality", "multimodal transportation" are reported all as "Multimodality"; "drone", "drone delivery", "UAV" are all condensed in "Drone"). In this way the fragmentation given by the non uniform use of keywords is reduced and trends and topics are more easily recognizable.

Table 3 reports the frequency of the major keywords considering papers published from 2015 only. Three considerations are worth to be noted. The first is that the keyword *Drone* is fully concentrated on papers published from 2015 on. Indeed, drone delivery is a contemporary topic that is gaining a lot of attention. The second is the recent focus on environmental impact of transportation, as witnessed by the frequency of the keyword *Emissions*. The third is the recent interest in *Synchromodality*, which is related to the technological advancements in information technology. These research directions will be further discussed in the rest of the paper. is

We classify the papers based on the modality used in the long-haul leg of the shipment and on the nature of the decision problem studied. Fig. 3 shows how the papers are divided among the modalities used in the long-haul leg.

We have mostly considered papers where a mathematical programming model is proposed. Table 4 provides a classification of the 104 papers (out of 118) proposing a mathematical program-

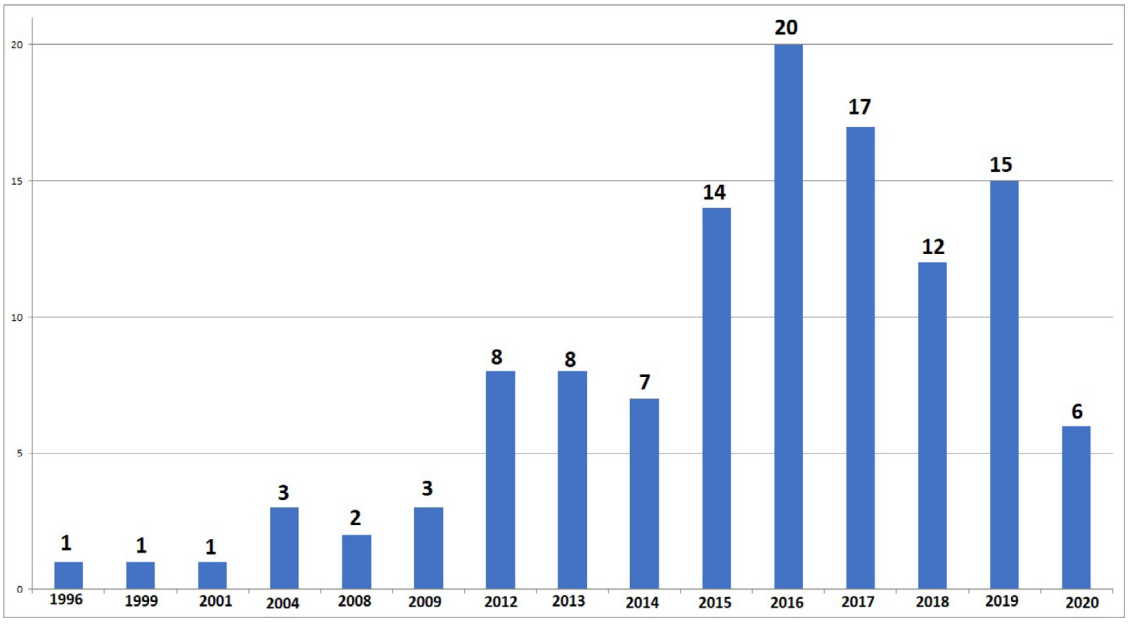


Fig. 2. Frequency by year of publication.

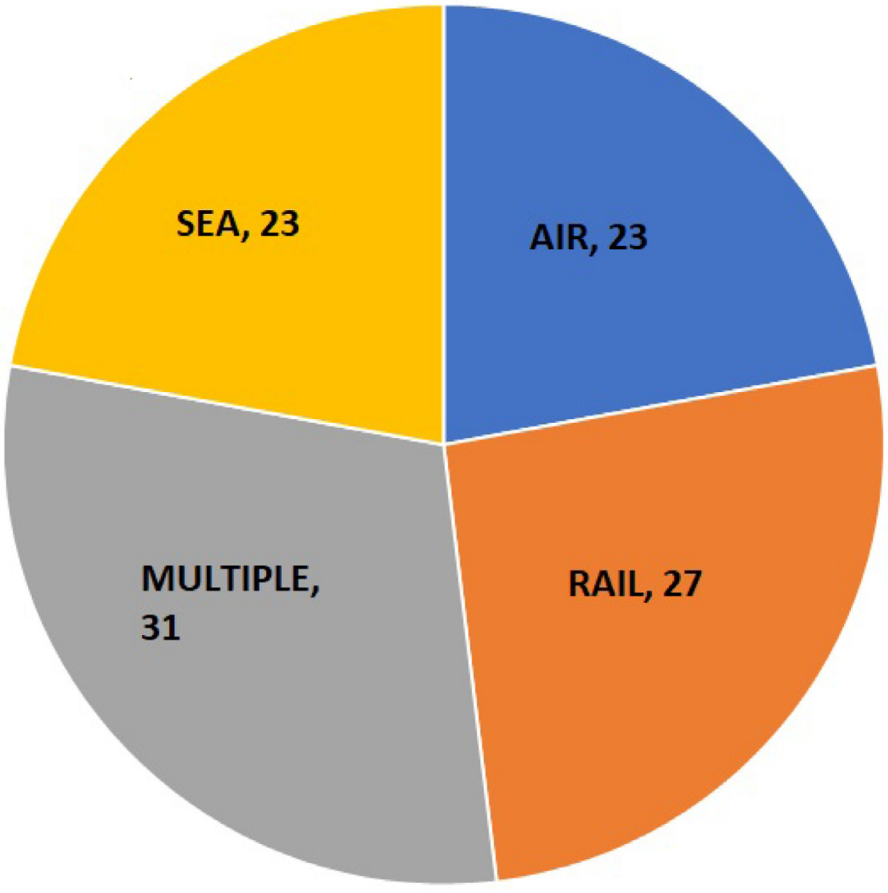


Fig. 3. Classification by modality on the long-haul leg.

ming model. The remaining papers present topics related to multimodal transportation, theoretical results that are useful for the solution of the most common formulations, reviews on specific topics and works of different nature which do not present optimization models but are nonetheless valuable for the scope of the literature review.

When a single modality is used on the long-haul leg, the most studied problems are the ones considering the multimodal transportation on rail (27 out of 104 papers with a mathematical programming problem). Modal shift from transportation carried out by trucks to a more sustainable option, using trains, is one of the main scopes of western and eastern countries policy makers, and

Table 4
Papers classification.

Modality	Operational	Tactical/Strategic	Total
Rail	7	20	27
Sea	10	14	23
Air	18	5	23
Multiple	9	23	31

therefore the popularity of research in this field is not surprising. Multimodal transportation making use of air services saw a recent surge in optimization research due to new technologies exploring the possibility of drone deliveries. Drone-delivery based problems are an exception to the logic used in this paper since the air modality is used in the end-haul *last-mile* delivery, and not in the long-haul leg. The majority of the problems studying drone delivery optimize the problem where a truck and a drone collaborate in order to carry out deliveries to customers. The drone can travel with the truck, depart from the truck to deliver a package to a customer and then return to the truck. The delivery route consists in parts where the truck and the drone travel separately, and others where the drone travels together with the truck. Problems where only drones are considered (e.g., a drone leaves the warehouse, delivers the goods and returns to the warehouse) are not considered since there is no use of other modalities. Truck-drone is a multimodal transportation and the related literature is reviewed in the Air section even though the transportation by air is in the end-haul part of the shipment. Problems studying optimization with more than two modalities are fairly recent, since 26 out of the total of 31 papers were published in the period 2015–2020. It is also the most studied category in this period (26 out of 75 papers, 18 papers are categorized as Air, 17 as Rail and 9 as Maritime).

Problems tackled by researchers in the field of multimodal transportation are very often inspired by real case applications, with 61.86% of papers containing real data or seeking optimization of real size instances in order to find solutions that can be effectively used in practice. Due to the complexity of these problems and the fact that in practice the solutions need to be found in limited amounts of time, the use of commercial solvers like CPLEX or GUROBI is not always viable. In fact, the majority of papers (72 out of 104, 69%) propose a heuristic approach.

A final note is about the journals on which the research on multimodal transportation has been published. Due to the nature of the field of research, the largest number of papers are published in operational research, computer science and transportation science focused journals (Transportation Research, European Journal of Operational Research, Transportation Science). On the other side, the number of publications in journals not entirely focused on logistics is noticeable, with journals interested in production, industrial engineering, business and management. Table 5 shows an overview of the journals where more than one reviewed paper appeared.

In Sections 3–5 we discuss problems involving one modality in the long-haul leg whereas Section 6 is dedicated to problems where two or more modalities are involved.

3. Air transportation

For years, air transportation was limited to the transportation of high valued material (according to the IATA, International Air Transport Association, in their "Value of Air Cargo" report, in 2016 air transportation moved 35% of the total in value). Recently, according to IATA, air transportation has grown in both relevance and volumes transported, thanks to e-commerce practices and the increased importance of Global Value Chains, as it is capable of guaranteeing fast delivery of products from and to the entire globe. Indeed, the transportation of freight through air services is often the only feasible option when tight constraints are imposed on the delivery time. Moreover, in the last lustrum, when most of the retail and distribution largest companies announced their drone delivery projects (Amazon, Walmart, DHL, UPS and Google are some examples), a great interest sparked to explore optimized solutions for last mile delivery using drones, technically known in the literature as Unmanned Aerial Vehicles (UAV).

Table 5
Papers classification.

Journal	N papers
Transportation Research Part E	13
European Journal of Operational Research	9
Transportation Research Part B	5
Transportation Science	5
International Journal of Production Economics	4
Computers & Industrial Engineering	4
Transportation Research Part C	4
Journal of Transport Geography	4
Computers & Operations Research	4
Procedia	3
Computers in Industry	2
Flexible Services and Manufacturing Journal	2
International Journal of Shipping and Transport Logistics	2
Optimization Letters	2
Omega	2
Computers in Industry	2
Transportation Research Record	2
Other (1 paper)	49
Total	118

3.1. Tactical/strategic problems

The literature on tactical or strategic multimodal transportation problems using the air modality is rather scarce. If we do not consider problems using more than two modalities (reviewed in Section 6), the only examples of tactical/strategic optimization problems involving the use of trucks and aircrafts is in Service Network Design (SND) applications. Specifically, an international courier needs to develop a dense network of air services and road pickup/delivery services in order to offer to its customers fast and reliable transportation schedules. Identifying hub locations and the services connecting hubs and collection/delivery areas is the main focus of SND problems, tackled in Barnhart & Schneur (1996) and Armacost, Barnhart, Ware, & Wilson (2004), where the authors introduce the concept of composite variables and use them to solve a set-covering formulation of the problem. In Yang (2009) the author split the problem in two subproblems: at first, the location of hubs is identified, then the flows running through them are optimized.

3.1.1. Air service network design

In this section we present a deeper analysis of the SND model introduced in Barnhart & Schneur (1996). This is the first work addressing this problem in the air transportation setting. The problem involves the use of a fleet of aircrafts, a fleet of vehicles (trucks/vans), a set of shipments that require to be served (either pickup or delivery), one main hub location and a set of Shipment Centers (SCs), linked together according to a hub-and-spoke layout. Both the main hub and the SCs are used as consolidation and shipping locations. Shipments are collected and transferred to the nearest SC where consolidation with other shipments is performed. Each shipment has an assigned SC of origin and a SC of destination. Transportation between SCs can be done by road transportation using feeder truck services, air transportation (using the owned fleet of different type of aircrafts, or using external commercial air services by acquiring space on the airfreight market) or by a combination of the two modalities. As it is not convenient to serve every SC directly with air services, aircrafts visit only a subset of

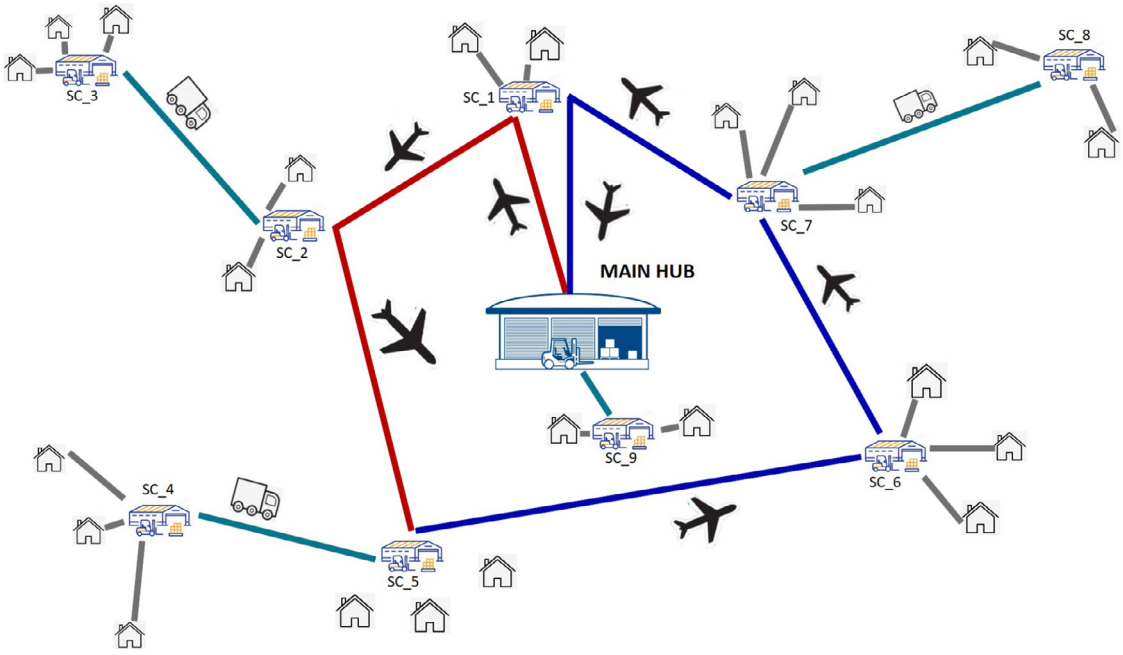


Fig. 4. Example of network structure.

SCs (air-stops). Each aircraft route is divided in a delivery route followed by a pickup route. From the air-stop SCs, either in the delivery or pickup route, a series of feeder road transportation services depart to other SCs not directly reached by air transportation. A visual representation of an example of the network structure is given in Fig. 4.

In the network represented in Fig. 4 there are 9 SCs. The delivery route is identified in red. It starts at the main hub and moves, in order, to SC_1, SC_2, SC_5. In SC_5 the delivery route ends and the pickup route begins. The pickup route is identified in blue. It starts in SC_5, visits SC_6, SC_7 SC_1 and ends at the main hub. SC_9 is served directly from the main hub, with either a truck or external commercial air service. Similarly, SC_3 is connected to SC_2, SC_4 is connected to SC_5 and SC_8 is connected to SC_7 by either truck services or external commercial air services. Moreover, time windows that need to be respected are assigned to both the hub and the SCs. For the SCs this comes in the form of *earliest pickup* and *latest delivery* times, imposed by the required level of service.

The model formulation makes use of the following sets:

- V is the set of owned aircraft types in the fleet;
- S is the set of SCs;
- F is the set of solutions in the feasible region, i.e., routes that satisfy all side constraints, such as stability, equipment balance, aircraft stop timing;
- D^v is the set of available delivery routes for aircraft type $v \in V$;
- U^v is the set of available pickup routes for aircraft type $v \in V$.

The model parameters are the following:

- c_l is the cost of serving route $l \in D^v \cup U^v$ with owned aircrafts. The cost includes all the costs associated with the delivery (pickup) route, i.e., the sum of air transportation costs, feeder ground transportation costs. If the quantity to be transported is greater than the capacity of the aicraft, the commercial cost of buying transportation services from third party air companies is included in c_l;
- c_i is the cost of serving all deliveries (pickups) of SC i using external commercial services;
- n_v is the number of aircrafts available for each type of aircrafts in the fleet;
- α_l^i is a parameter equal to 1 if route l serves the pickups of SC i, 0 otherwise;
- δ_l^i is a parameter equal to 1 if route l serves the deliveries of SC i, 0 otherwise;
- γ_l^i is a parameter equal to 1 if the first (last) air-stop on pickup (delivery) route l is SC i.

The formulation makes use of the following variables:

- $\hat{p}_i$, is a binary variable that is equal to 1 if all pickups of SC i are served by commercial air, and 0 if SC i is part of a route of an owned aircraft;
- $\hat{d}_i$, is a binary variables that is equal to 1 if all deliveries of SC i are served by commercial air, and 0 if SC i is part of a route of an owned aircraft;
- x_l is a binary variable equal to 1 if route $l \in U^v \cup D^v$ is flown, 0 otherwise. Pickup (delivery) routes are defined by a set of pickup (delivery) stops, the order in which stops are visited, the ground feeder services that are needed to fulfil pickups (deliveries), the time at which the route leaves (arrives) at the hub and at the SCs. Routes are either pickup or delivery cycles. They begin or end at the hub, but are not complete tours. Specifically, a delivery route starts from the hub but does not end at the hub and a pickup route ends at the hub but does not start from the hub. Thus, pickup and delivery routes need to be combined together in order to form complete cycles from and to the hub.

The formulation of the problem is as follows:

$$\min \sum_{v \in V} \sum_{l \in U^v \cup D^v} c_l x_l + \sum_{i \in S} c_i \hat{p}_i + \sum_{i \in S} c_i \hat{d}_i \tag{1a}$$

$$\sum_{v \in V} \sum_{l \in U^v} x_l \alpha_l^i + \hat{p}_i = 1 \qquad i \in S \tag{1b}$$

$$\sum_{v \in V} \sum_{l \in D^v} x_l \delta_l^i + \hat{d}_i = 1 \qquad i \in S \tag{1c}$$

$$\sum_{l \in D^v} x_l \gamma_l^i - \sum_{l \in U^v} x_l \gamma_l^i = 0 \qquad i \in S, v \in V \tag{1d}$$

$$\sum_{l \in U^v} x_l \leq n^v \qquad v \in V \tag{1e}$$

$$x_l \in F \tag{1f}$$

$$x_r, \hat{p}_i, \hat{d}_i \in \{0,1\} \qquad i \in S, l \in U^v \cup D^v, v \in V. \tag{1g}$$

The objective function (1a) minimizes the total cost given by the cost of the routes selected and the commercial services used. Constraints (1b) and (1c) ensure that all shipments are picked up and delivered. Constraints (1d) pair delivery and pickup routes in order to create complete tours starting and ending at the hub. Constraints (1e) limit the maximum number of aircrafts used to those available for each type. Constraints set (1f) contains all the technical and operational constraints defining the feasible routes. Constraints (1g) force the variables to assume either value 1 or 0. Note that this formulation is flexible enough to handle different problem settings, by properly defining set F. For example, set F may include time window constraints on the hub or on the SCs or additional side constraints like incompatibilities between SCs and specific types of aircrafts. (at

The formulation requires the generation of all the feasible aircraft routes. The combinatorial magnitude of the task is massive and by considering even a small set of SCs the number of feasible routes becomes huge. Moreover, the possibility to externalize transportation using commercial services increases the potential size of the model. The formulation is thus solved through column generation where the subproblem includes all the constraints on route feasibility, such as constraints on air-stops, ground feeders services, fixed schedules or time windows.

This formulation is the basis of successive works such as Armacost et al. (2004) where the authors deal with a similar problem by using *composite variables*. Composite variables contain information regarding the number of aircrafts used and the capacity available. The formulation provides a much stronger linear relaxation than the one of formulation (1a)–(1g). A SND formulation is also used in Yang (2009). Due to the stochastic nature of the problem studied, the resulting formulation is a two-stage stochastic programming problem. reduced

3.1.2. Trends and future research

Problems involving a combination of air transportation with either rail or truck have been rarely explored in the literature, apart from SND problems. Hub optimization problems are studied in Saberi & Mahmassani (2013), Alibeyg, Contreras, & Fernández (2018) and Yang, Yu, & Chen (2016), but multimodality is never considered and environmental aspects are rarely addressed. A review of the problems related to air transportation treated in the literature can be found in Ginieis, Sánchez-Rebull, & Campa-Planas (2012). We note that, from the pioneering studies mentioned in this section, not many developments were made in service network design problems involving air transportation, especially not in networks where air transportation is coordinated with other modalities. Integration between air and maritime transportation seems unlikely, as both the modalities are used for international, long distance trades and the characteristics of one modality are the opposite of the other (maritime is a slow, low cost transportation while air is an extremely fast and expensive option), making them mutually exclusive. On the other side, networks integrating rail and air transportation are worth being studied. Right now, most of the air companies move their goods from hub to hub (airports), if positioned in the same region, mostly by truck services. Creating a rail network connecting airports with other airports or collection areas could greatly improve the quality of service and the environmental impact, and it is worth studying. In addition, network design problems aim at minimizing either transportation cost or transportation time. Multi-objective problems should be further investigated possibly including, whenever possible, environmental aspects such as carbon emissions.

3.2. Operational problems

Operational problems in recent years mostly focused on exploring optimal solutions and strategies to effectively exploit the use of drones in last mile deliveries. This is a recent trend of research that sparked great interest, and even though in this case air transportation is not used in the long-haul leg, but in the end-haul leg to handle deliveries, the cooperation between truck and drones is a good example of the use of multimodal transportation. Since its introduction in Murray & Chu (2015), the Flying Sidekick Traveling Salesman Problem (FSTSP) received much attention by researchers. The problem consists in identifying the best route to be followed by trucks and drones, collaborating in order to optimally deliver small parcels to customers, with the objective of minimizing the delivery time. Developments can be found in Carlsson & Song (2017), where the authors use continuous approximation methods and are able to demonstrate that the improvement in delivery time is proportional to the square root of the ratio of the speeds of truck and UAV. Dell'Amico, Montemanni, & Novellani (2019) present valid inequalities to be used in FSTSP formulations. The Traveling Salesman Problem with Drones (TSP-D) explores the collaboration between trucks and drones to carry out deliveries to final customers, with minor differences with respect to FSTSP. Poikonen, Golden, & Wasil (2019) introduce a tailored branch-and-bound algorithm for the TSP-D. Kitjacharoenchai et al. (2019) and Agatz, Bouman, & Schmidt (2018) propose models where the objective is the minimization of delivery times, while Ha, Deville, Pham, & Hà (2018) and Ha, Deville, Pham, & Hà (2020) explore the same problem but seek the minimization of costs. Multi-vehicle cases are studied in Wang & Sheu (2019) and in Poikonen, Wang, & Golden (2017), where the Vehicle Routing Problem with Drones (VRPD) is presented. Wang, Poikonen, & Golden (2017b) analyze various VRPD scenarios and present a worst-case analysis. Di Puglia Pugliese & Guerriero (2017) study the last-mile delivery problem with time windows using drones. The use of drone stations is studied in Kim & Moon (2018) and Ferrandez, Harbison, Weber, Sturges, & Rich (2016).

Apart from UAV deliveries, operational problems in air transportation are scarcely studied. Recently, Archetti & Peirano (2020) and Angelelli, Archetti, & Peirano (2020) studied the optimization problem tackled by freight forwarders who need to handle air transportation shipments. Ruan, Wang, Chan, & Shi (2016) optimize the intermodal transportation of medical supplies proposing a stochastic model. Often, the areas struck by natural calamities are inaccessible with normal means of transportation and, thus, the integrated use of trucks and helicopters is needed. An interesting overview of humanitarian logistics, often making use of helicopters, can be found in Ertem, Melike, & Arslan (2017).

3.2.1. Traveling salesman problem with drones

Traveling Salesman Problem with Drones (TSP-D) saw a spark of interest in the last few years and different formulations have been proposed. Here, we present the nice and compact formulation for the TSP-D introduced by Agatz et al. (2018). There exists previous papers proposing mathematical formulations for the TSP-D (the first to propose a formulation for such a problem is Murray & Chu (2015)). We have chosen the one in Agatz et al. (2018) as it is a general formulation that can be adapted to specific variants of the problem.

The TSP-D is modeled on a graph $G = (N, A)$ where node i_0 is the depot and nodes $i_1, i_2, \ldots, i_n$ are the customer locations. $c(a) = c(i, j)$ and $c^d(a) = c^d(i, j)$ are the travel time from node i to node j for truck and drone, respectively. It is assumed that a drone is always faster than a truck, thus $c^d(a) \leq c(a)$. The objective of the TSP-D is to find the route with the shortest time needed to serve all customers using truck and drone. Customers can be served by both types of vehicles (truck and drone) when possible, otherwise they are inserted in the subset $N^t \subseteq N$ of customer nodes that can be served by truck only. A few assumptions are made. First, the drone can deliver only one package at a time, after which it needs to return to the truck (or to the depot). Moreover, the drone can rendezvous or depart with a truck at customer location or depot only (i.e. it cannot meet or separate along the way). Lastly, departure and landing times are assumed to be negligible and are not considered. The same assumption is made for the battery charging process. A solution of the TSP-D is given by the truck route $T = (t_0 = i_0, t_1, t_2, \ldots, t_l = i_0)$ combined with the drone route $D = (d_0 = i_0, d_1, d_2, \ldots, d_m = i_0)$ where l is the number of customers visited in the truck route and m is the length of the drone route. Clearly, each node contained in N^t must be assigned to T. The drone route D contains all nodes served by the drone, plus the nodes serving as rendezvous or depart locations. Thus, a distinction between nodes is needed. *Drone nodes* are nodes served by the drone. Similarly, *truck nodes* are nodes served by the truck. *Combined nodes* are nodes visited by both truck and drone. This implies that, once separated, truck and drone can serve customers separately and rejoin at different nodes. Computing the time needed to handle routes T and D is not enough though, since at combined nodes they need to join and, unless they are perfectly synchronized, one will need to wait for the other. Thus, in order to properly compute service time, Agatz et al. (2018) introduce the concept of operation. An *operation* o consists of two combined nodes, called the start node i and the end node j, at most one drone node, and a number of truck nodes. There are two kinds of operations. In the first one, the truck travels from the start node to the end node while the drone remains parked upon it. In the second one, the drone leaves the truck at the start node, delivers the package to a drone node and returns to the truck at the end node. The operation can be described as a pair of subtours $(\tilde{T}, \tilde{D})$ where $\tilde{T}$ is the sequence of nodes of the tour T that begins at the combined node i and ends at the combined node j, including all the truck nodes in the middle. $\tilde{D}$ is defined similarly, with the difference that only one drone node can be present between two combined nodes. The driving time of $\tilde{T}$ is defined as $c(\tilde{T}) = \sum_{a \in \tilde{T}} c(a)$ while the flying time of $\tilde{D}$ is defined as $c(\tilde{D}) = \sum_{a \in \tilde{D}} c^d(a)$. An operation o is a combination of a route $\tilde{T}$ and $\tilde{D}$, and the time needed to perform each operation o is $c(o) = \max\{c(\tilde{T}), c^d(\tilde{D})\}$.

The model formulation makes use of the following sets:

- O is the set of feasible operations. The basic version of the model requires, as previously explained, the presence of two combined nodes. Limits on battery charge or any other operational constraint can be added to further define the feasible region for operations;
- $O(i)$ is the set of all operations containing node i;
- $O^-(i) \subset O$ is the set of operations that use node i as starting node;
- $O^+(i)$ is the set of operations that use node i as ending node;
- $O^-(S)$ is the set of operations that use one node in subset $S \subset N$ as starting node and one node in $N \setminus S$ as ending node. Similarly, $O^+(S)$ is the set of operations that use one node in subset S as ending node, and start from a starting node in $N \setminus S$.

The decision variables are:

- x_o is a binary variable indicating whether operation o is selected or not;
- z_i is a binary variable indicating if node i is chosen as starting node in at least one operation.

The formulation proposed is the following:

$$\min \sum_{o \in O} c_o x_o \tag{2a}$$

$$\sum_{o \in O(i)} x_o \geq 1 \qquad i \in N \tag{2b}$$

$$\sum_{o \in O^-(i)} x_o \leq z_i \qquad i \in N \tag{2c}$$

$$\sum_{o \in O^+(i)} x_o = \sum_{o \in O^-(i)} x_o \qquad i \in N \tag{2d}$$

$$\sum_{o \in O^+(S)} x_o \geq z_i \qquad S \subset N \setminus \{i_0\}, i \in S \tag{2e}$$

$$\sum_{o \in O^+(i_0)} x_o \geq 1 \tag{2f}$$

$$z_{i_0} = 1 \tag{2g}$$

$$x_0 \in \{0, 1\} \qquad o \in O \tag{2h}$$

$$z_i \in \{0, 1\} \qquad i \in N \tag{2i}$$

The objective function (2a) minimizes the time needed to serve all customers. Constraints (2b) ensure that every node is covered by at least one operation. Constraints (2c) force z_i to take value 1 if at least one operation uses i as starting point. Constraints (2e) and (2f) ensure connection and constraints (2d) are degree constraints. A route is always forced to start from the depot by constraint (2g). Finally, (2h) and (2i) define variables domain.

In Agatz et al. (2018) a great deal of theoretical insights are provided. We report here two properties of optimal solutions. First, an optimal solution of the TSP-D could require that combined nodes are visited more than once. Secondly, optimal solutions require that both truck and drone nodes are visited exactly once. These properties are the base of the development of the heuristic approaches proposed in the paper. The formulation proposed by Agatz et al. (2018) can be easily adapted to handle practical settings. Similar formulations are presented in Poikonen et al. (2019) and Kitjacharoenchai et al. (2019).

3.2.2. Trends and future research

The use of UAV is the prominent trend in intermodal air transportation literature. The field is new and completely missing from the literature review in SteadieSeifi et al. (2014). The numerous studies, though, do not consider real data instances and purely rely on randomly generated scenarios. Recently, Jeong, Song, & Lee (2019) explored real-world feasible solutions, considering both payload-energy dependency and the activation of no fly zones. Solutions are found using a two-phase constructive heuristic. However, the authors applied the model to the randomly generated benchmark instances presented in Murray & Chu (2015) and not to real data. It would be of great interest to study real applications to verify whether the solution found generates valuable improvements in real life applications. Generally speaking, researchers should expand the formulations by introducing constraints inspired

by limitations that could be applied to real world applications. Other than problems combining truck and drones for last-mile deliveries, the research in multimodal operational problems where the long-haul leg of the transportation is carried out by air is fairly scarce. No studies of operational problems involving air transportation were presented in SteadieSeifi et al. (2014) and it seems this field was ignored up to this date, apart from the few contributions mentioned above. However, this is a potential fertile ground for research, given the extremely reduced delivery times that are nowadays offered by selling companies, especially those involved in e-commerce. In addition, dynamic optimization of commodities flow using air transportation in the long-haul is a potential objective of synchromodal optimization problems.

4. Rail transportation

The theme of multimodal freight transportation, featuring the long-haul leg carried out by means of rail transportation, is an extremely contemporaneous and deeply studied topic. The urge of rethinking the freight transportation network and the goal of reducing emissions is on top of policy makers agenda for well developed countries. In the White Paper 2011 (Commission (2011)), in point 6, the European Commission states *the need to drastically reduce world greenhouse gas (GHG) emissions, with the goal of limiting climate change below 2° C. Overall, the EU needs to reduce emissions by 80–95% below 1990 levels by 2050, in the context of the necessary reductions of the developed countries as a group, in order to reach this goal*. Since the transportation industry is one of the main contributors to the emissions of GHG, a shift of the modal split in favor of rail is required.

The main drawback of multimodal rail-road transportation, when compared to unimodal road transportation, is that it requires a transshipment multimodal terminal where goods are moved from the road vehicles, usually trucks, to cargo trains. This operation not only is time consuming and generates costs, but impacts negatively on the transit and delivery times when terminal operations are not carried out efficiently, with negative effects on the service level offered to customers.

Out of the 27 studies that optimize the multimodal transportation using road and rail services, 7 tackle problems of operational nature while 20 seek optimization for tactical and/or strategic problems. In addition, two papers propose a cost analysis which is helpful in understanding the cost structure of intermodality between rail and road. A model which analyzes the generalized cost of intermodal transportation by considering the multiple aspects that contribute to its amount has been proposed by Hanssen, Mathisen, & Jørgensen (2012). It is common knowledge that, due to its cost structure (higher fixed cost, infrastructural expenses and transshipment costs), intermodal transportation is more convenient with respect to an unimodal solution only if the total distance, in particular the segment of the long-haul part of the transportation service, is higher than a certain amount, and predominant with respect to the total distance from origin to destination. In their work, the authors claim that the optimal long-haul transport distance in which intermodal transportation becomes preferable, when compared to unimodal road transportation, increases if: handling costs at terminal increase, total transportation distance increases (unimodal transportation usually has a higher kilometric cost, therefore the longer is the distance to be covered the more likely intermodality will be a better choice), pre- and post- haul costs increase, marginal costs for rail increase and marginal costs for truck decrease. As a general rule, the higher is the impact of intermodal costs (handling, transshipment, pre- and post- haul) or the lower is the cost of the unimodal solution, the longer needs to be the long-haul distance in order to achieve convenience for multimodality. Another study on intermodal costs is Kordnejad (2014), where the author presents a model of a regional rail based intermodal transport system in the Stockholm region. The model considers a great deal of aspects for each component contributing to the calculation of costs such as rail operations (number of locomotives and wagons, weight, emission factor, insurance and capital cost among the others), road haulage (average speed, maintenance, driver costs, interest rate) and terminal handling (infrastructural costs, shunting, transloading operators and equipment). The results indicate that the most critical parameters for such a system to be competitive are the loading space utilization of the train and the cost for terminal handling.

4.1. Tactical/strategic problems

Tactical/strategic problems mainly differ for the time span considered for the decision. For instance, a problem has a tactical nature if the objective is to identify a transshipment terminal with which to sign medium or long-term contracts, and strategic if the objective is to build a transshipment terminal in order to operate it for a longer period of time. In both cases the problem is to identify the location of the facility. Location problems for intermodal terminals are a prominent topic, especially since a major policy maker objective is the transfer of traffic from trucks to trains in order to reduce both emissions and congestion. Examples can be found in Limbourg & Jourquin (2009) where a Non Linear-MIP model is presented in order to solve a p-hub median problem and build a hub-and-spoke network, and in Sörensen, Vanovermeire, & Busschaert (2012) who propose a MIP model for the intermodal terminal location problem, and solve it with the use of two heuristics making use of a two-phase structure. Both heuristics start with a construction procedure (GRASP and ABHC) and explore possible improvement through the use of local searches. The model aims at improving the efficiency of intermodal transportation by enhancing the connectivity between terminals. Lin & Lin (2016) expand on the work of Sörensen et al. (2012) by refining the model and presenting a Binary Linear Program (BLP) model solved through the use of a two-stage matheuristic based on relaxing variables and adding constraints to the model. Examples of dry ports location problem following single hub itineraries are Ambrosino & Sciomachen (2014), where a mathematical programming model based on a two-level weighted multimodal graph is proposed which is applied to a real world application in northern Italy, and Ambrosino & Sciomachen (2016), where innovative features for modelling capacity bounds on the hubs are presented and the multimodal capacitated hub location problem is introduced. One noticeable aspect is that the number of hubs to be located is not a fixed parameter as in the classical p-hub median problem, but is a decision variable. Moreover, split deliveries are allowed, while in intermodal location problems this is typically not the case.

Intermodal transportation inherently needs efficient logistics networks. The optimal design of both physical networks (i.e. the creation of railway, transshipment terminals and hubs) and service networks (a set of connected rail and road services) is a widely studied topic. Interesting applications of network design can be found in Tonneau et al. (2015), where the optimization of a waste supply chain, involving rail transportation, is studied.

In Crainic (2000) the author gives an overview of basic models, evolutions and solution methods. Arnold, Peeters, & Thomas (2004) propose a linear binary formulation based on multi-commodity fixed-charge network design problems and solve it through heuristics. In Zhang, Wiegmans, & Tavasszy (2013) a non-linear model for the optimization of multimodal terminal network design is proposed. The authors include the cost of carbon emissions and the effects of economies of scale at terminals. The proposed solution approach is a genetic algorithm. Karimi & Bashiri (2018) propose a multi-commodity multimodal supply chain net-

work which is designed for intelligent manufacturing. The objective is to identify the location of the hubs where flows are split between rail and road transportation. A mixed integer linear programming model is proposed to minimize the total logistics costs. Poudel, Marufuzzaman, & Bian (2016) study the design of a multimodal transportation network for biomass transportation, proving that an extended use of the multimodal option can efficiently deal with supply uncertainties. Fotuhi & Huynh (2018) propose a robust mathematical model for the network expansion problem, solving different instances in different time periods, each one influencing the successive time periods. The solution is found using a hybrid simulated annealing algorithm.

Flow formulations are also used in problems with tactical and strategic nature. Often, the study of the flows in the network can help in identifying improvements to it, or in understanding the effects and impacts of changes. Examples of flow formulations used to optimize tactical decisions can be found in Ambrosino, Ferrari, Sciomachen, & Tei (2016), where a MILP formulation is presented in order to optimize the flow of goods moving through the network while identifying the position of the intermodal terminals. Different networks inspired by real world instances are compared and managerial conclusions are drawn, considering the impact of external costs. In Ambrosino & Sciomachen (2017) the authors present a study where the problem is to identify the best set of investments, in order to optimize the increase of modal split favouring rail over road through the use of a MILP formulation which aims at optimizing the flow of goods in the network and the amount of investments to be done. Uddin & Huynh (2015) seek an a-priori optimal assignment of traffic over rail and road, in a large scale intermodal network. They propose a model with a non linear objective function and linear constraints and use a gradient projection algorithm to find the optimal solution. A similar idea is used to decide how to redirect traffic in view of disruptive events in Uddin & Huynh (2016), where a non-linear programming model is approximated with a MILP formulation using a sample average approximation algorithm. The aim of this work is to find optimal routes for shipments in a network under disruption. Experiments are conducted based on a real network in the US and the results show that an intermodal solution is faster, less expensive and more robust when compared to unimodal solutions currently used by freight operators.

When the choice of the best way to move shipments through a network directly influences the design of the network itself and, at the same time, the design of the network influences the selection of the services used to ship cargo through it, then the problem studied is the so called Service Network Design (SND). Ishfaq & Sox (2012) propose a SND problem that considers delays at hubs and uses tabu search to solve it, while Bouchery & Fransoo (2015) include environmental aspects in the form of carbon emissions and externality costs in their non linear model. Their service network design problem is optimized based on two different measures of distance, namely Manhattan and Euclidean. Moreover, the authors prove that, in order to optimize carbon emissions, an optimal level of modal split needs to be identified, as otherwise no improvements and even worse solutions can be found. Laaziz & Sbihi (2019) study the problem of a freight forwarder that needs to optimize a rail-road network in order to efficiently handle a large number of shipments. No real data is used but the MILP model solves efficiently real size instances.

4.1.1. Network design

Network Design (ND) is one of the most studied problems in intermodal transportation where the long-haul leg makes use of rail services. Inspired by Crainic (2000), we will present in this section an introduction to the problem and two general formulations. The two formulations are the so-called *linear cost, capacitated, multi-commodity network design* (MCND) version and the *path-based* MCND version of the problem. While the problem is the same, the two formulations have different characteristics and both are widely used. Path-based formulations in particular are more suited for column generation approaches and matheuristics.

ND problems are used in intermodal transportation in order to take optimal decisions regarding, for instance, the construction of facilities such as terminals for transshipment, the selection of the best set of transportation services to be offered, the best allocation of resources for investments. ND formulations make use of graphs defined by nodes connected usually by direct links (arcs). Nodes can represent any location of interest such as origins, destinations, transshipment facilities. Arcs are usually associated with fixed costs (costs that are generated at the activation of the connection, for instance the cost of building a railroad) and variable costs (costs that depend on the use of the arc, for instance, the more traffic the railroad sustain, the more maintenance costs will be encountered). Multiple characteristics can be assigned to both arcs and nodes.

The typical objective of the problem is to select the arcs of the network that allow the satisfaction of the transportation demand, while minimizing the total cost which is given by fixed and variable costs. We are given graph $G = (N, A)$ where N is the set of nodes and A is the set of arcs of a network in which a set K of commodities are transported. Nodes are used to represents locations while arcs represent connections between locations. Connections can be either physical (roads, rails) or transportation services. Let us define the following parameters:

- $\bar{c}_{ij}$ is the unitary fixed cost associated with each activation of arc (i, j);
- c^k_{ij} is the transportation cost for each unit of flow of commodity k traversing arc (i, j);
- u_{ij} is the capacity of arc (i, j);
- d^k_i is the demand of commodity k at node i.

The formulation makes use of the following variables:

- $y_{ij}, (i, j) \in A,$ are integer variables that are used to model discrete design decisions. If $y \in \{0, 1\}$ the choice is limited to the activation of the arc, while if $y \in N_+$ the variable is used to represent the quantity of facilities to build or the capacity of the arc or the level of service offered;
- $f^k_{ij}, (i, j) \in A, k \in K,$ are continuous flow variables representing the amount of traffic of commodity k transiting through the arc (i, j).

In case y_{ij} variables are binary, the MCND formulation is the following:

$$\min \sum_{(i,j)\in A} \bar{c}_{ij} y_{ij} + \sum_{(i,j)\in A, k\in K} c^k_{ij} f^k_{ij} \tag{3a}$$

$$\sum_{j\in N} f^k_{ij} - \sum_{j\in N} f^k_{ji} = d^k_i \qquad i \in N, k \in K \tag{3b}$$

$$\sum_{k\in K} f^k_{ij} \le u_{ij} y_{ij} \qquad (i, j) \in A \tag{3c}$$

$$f^k_{ij} \ge 0 \qquad (i, j) \in A, k \in K \tag{3d}$$

$$y_{ij} \in \{0, 1\} \qquad (i, j) \in A \tag{3e}$$

The objective function (3a) seeks the minimization of the total cost. Constraints (3b) balance the flow in each node and ensure the satisfaction of the demand while constraints (3c) limit the

flow in each arc within the available capacity. This formulation is widely used and examples of studies expanding from this basic formulation are Limbourg & Jourquin (2009), Sörensen et al. (2012), Ambrosino & Sciomachen (2014), Ambrosino & Sciomachen (2016), Tonneau et al. (2015), Karimi & Bashiri (2018), Ishfaq & Sox (2012). limited

An equivalent *path-based* MCND formulation has been proposed by the same author. The formulation uses the definition of set L^k which is the set of feasible paths for commodity $k \in K$.

The parameters are:

- δ_{ij}^{lk}, is a parameter indicating whether arc (i, j) is included in path l for commodity k;
- c_l^k is the transportation cost for each unit of commodity k along path l.

The demand d_i^k is aggregated into d^k which is the total demand of commodity k. The demand of each node is taken into account when generating the feasible paths in L^k. In addition to the variables y_{ij} defined above, the path-based MCND formulation uses the following variables:

- $f_l^k, l \in L^k, k \in K$, is a variable representing the flow of commodity k along path l.

The formulation is the following:

$$\min \sum_{(i,j)\in A} \bar{c}_{ij} y_{ij} + \sum_{k\in K} \sum_{l\in L^k} c_l^k f_l^k \tag{4a}$$

$$\sum_{l\in L^k} f_l^k = d^k \qquad k \in K \tag{4b}$$

$$\sum_{k\in K} \sum_{l\in L^k} f_l^k \delta_{ij}^{lk} \le u_{ij} y_{ij} \qquad (i,j) \in A \tag{4c}$$

$$f_l^k \ge \qquad l \in L^k, k \in K \tag{4d}$$

$$y_{ij} \in \{0,1\} \qquad (i,j) \in A \tag{4e}$$

Path-based formulations are used in Fotuhi & Huynh (2018), Lin & Lin (2016), Arnold et al. (2004), Uddin & Huynh (2015), Laaziz & Sbihi (2019). The path-based formulation is tailored for the use of column generation approaches, as depicted in Crainic (2000). Different solution approaches, as cutting planes or branch-and-bound, are often used to solve network design problems. References and examples can be found in Crainic (2000). When dealing with practical applications, as the complexity and the size of the problem increase, exact algorithms are not suited. Heuristic approaches such as tabu search, simulated annealing and genetic algorithms are often adopted. Again, references and examples can be found in Crainic (2000) and also in Lin & Lin (2016) which expands on the path-based formulation presented above.

4.1.2. Trends and future research

Tactical and strategic problems involving multimodality with a combination of road and rail transportation are fairly common and one of the first and most studied topics in the literature. Western developed countries intensified their efforts in shifting the modal split from the current situation, where road transportation is prominent, to a more sustainable multimodal solution. Henceforth, more and more studies consider environmental aspects, mainly in the form of carbon emissions. Other forms of negative externalities are not that prominent though, and future research should investigate efficient ways of handling transportation while effectively reducing, for instance, congestion or different forms of pollution such as acoustic pollution. SteadieSeifi et al. (2014) pointed out the lack of studies addressing transshipment costs and yard capacity. From 2014 multiple authors (Tonneau et al. (2015), Karimi & Bashiri (2018), Ambrosino & Sciomachen (2016), are some examples) addressed the issue and considered constraints and objective functions reflecting these characteristics of the problem. This trend should be fostered and expanded in the future.

Other directions of study with high potential are related to the development of combined systems with coordinated rail and road services. This would mean to enlarge the above-mentioned studies on SND to include coordinated road services. Such a coordinated system should be able to efficiently diminish carbon emissions and offer better and cheaper logistic services to customers. Innovative forms of distribution using intermodality, with an increased use of rail, can be explored. For instance, intermodal terminals may be located close to urban areas and equipped with anything necessary to offer warehousing and distribution services in order to reduce the length of pre- and end- haul transportation.

Also, the recent pandemic showed the importance of designing multimodal networks that are resilient against disruptions, especially considering situation when movements are limited not by physical damages to the network but by different causes. The works of Fotuhi & Huynh (2018) and Uddin & Huynh (2016), among the others, should be expanded. The results, where the service routes are proved to be more robust and are more suited to face disruption events with respect to the ones used by carriers, show that the use of intermodality between rail and road is a far superior option compared to the unimodal road transportation. Indeed, SteadieSeifi et al. (2014) pointed out the lack of research in robust network design.

4.2. Operational problems

Operational problems focus on the management of the day to day business. In particular, most of the operational problems aim to tackle issues arising during the daily management of operations. The most common problem tackled is the selection of the best way to move goods through a multimodal network where passing through transshipment terminals is mandatory to switch modality from road to rail. Optimization of network flows has been explored in Bierwirth, Kirschstein, & Meisel (2012), where the authors study the effect of consolidating different shipments in order to achieve a reduction of costs and an improvement of the level of service, whle proposing a MILP formulation of their model. A real-world case study is presented. Bhattacharya, Kumar, Tiwari, & Talluri (2014) propose an optimization model based on arc flows in a time-space network for the same problem. Lee, Oduor, Farahmand, & Tolliver (2014) present a linear programming model coupled with a geospatial dynamic trip assignment model. The objective of the study is to optimize an intermodal transportation flow by minimizing the total impedance given by the sum of transportation, inventory and transshipping impedances. Assadipour, Ke, & Verma (2015) study a shipment plan problem for hazardous material with a nonlinear formulation. They consider the congestion at intermodal terminals and aim to find the proper equipment capacity. They develop a genetic algorithm and they apply it to a real-size problem. The cited contributions on operational problems often make use of flow formulations. Nossack & Pesch (2013) propose a scheduling problem arising in intermodal transportation, where services of different modalities are coordinated in order to reduce waiting time at transshipment terminals and avoid congestion.

4.2.1. Trends and future research

Operational problems are less studied than tactical and strategic problems, and the majority of them are treated as flow optimization problems. Some works, as Kozanidis (2017) and

Dayarian, Crainic, Gendreau, & Rei (2015), consider vehicle routing problems (VRP) applied to pre-haul leg and end-haul leg. However, they are not combined with the long-haul leg transportation. Thus, it would be worth studying if and how VRP and flow optimization change in a combined environment where the flow decisions taken in the whole network are influenced by the solution of the VRPs at both pre-haul and end-haul legs. Also, alternative formulations other than flow formulations are worth being explored. Intermodal transportation is characterized by the typical network configuration and therefore flow formulations are a natural representation of the problem. However, different aspects with respect to the movement of commodities throughout the network, like service scheduling to achieve better synchronization, or shipment planning problems, should be explored, especially considering an integrated network setting. The work of Assadipour et al. (2015) is a nice example of implementing multiple aspect when considering the flow of goods throughout a network and the general idea presented should be expanded. A trend of the recent literature is the effort spent to tackle yard capacity and yard congestion issues and include transshipment cost. In this sense more integrated systems could also consider problems related to transshipment and transfer points, e.g., load and unload planning operations, traffic management at transshipment terminal coordinated with operation scheduling and yard management problems. Bhattacharya et al. (2014) put the emphasis on the importance of collaborative logistics and, as pointed out in SteadieSeifi et al. (2014), of the collaboration between different actors, and claim that the optimization of the whole logistic system is a goal to seek. Studies considering the idea of synchromodality aim, among other objectives, at helping cooperation between different actors.

The number of multi-objective and multi-commodity problems increased since the literature review in SteadieSeifi et al. (2014). The multi-objective aspect is related to the recent raise of environmental concerns. The multi-commodity feature is instead associated with the definition of specific constraints associated with the transported goods, which makes the corresponding problems more challenging.

5. Long-Haul Maritime and Waterway Transportation

According to Ebeling (2009), roughly 90%, in weight, of non-bulk global trade is carried out by sea. Maritime transportation is economic and permits the transportation of high quantities of raw materials like metal ore, wood, grain; or liquid bulk such as crude and refined oil, liquified gas, chemicals of any kind. Transportation of special material, particularly over-dimensioned or over-weighted goods, is often possible with maritime transportation only, and in any case it is the least expensive option. Containerized transportation is in constant growth (in 2018 according to the UNCTAD annual report 2019 it registered a growth of +2.6%) and it is the main actor in intermodal transportation. In 2018 the total container throughput of all ports of the world summed up to 162 millions TEUs (UNCTAD annual report 2019), with the top three ports in the world (Shanghai, Singapore, Ningbo-Zhoshuan) accounting for 105 millions TEUs and over 1.7 billion tons moved. International trade is inherently imbalanced, with some countries that are natural exporters of goods while others are natural importers. This is particularly evident in maritime transportation. The problem of containers return logistics is prominent and is driven by the need to move empty containers from import areas back to export areas. Optimization techniques have the potential of being helpful in finding cost effective solutions for managing this reverse flow. Motorways of the sea as an alternative to road transportation is still a discussed topic and an interesting field of research (a recent quantitative analysis is presented in Lupi, Farina, Orsi, & Pratelli (2017)). The concept was introduced with the 2001 Transport White Paper-European transport policy for 2010: time to decide. The European Commission proposed the development of Motorways of the Sea as a "real competitive alternative to land transport". The Trans European Transport Network (TEN-T) policy guidelines addresses the implementation and development of a Europe-wide network of railway lines, roads, inland waterways, maritime shipping routes, ports, airports and railroad terminals. In there, four main corridors were designed in order to decrease the traffic on road, moving vehicles and goods through ro-ro (roll on-roll off, where the whole vehicle given by tractor and trailer is loaded on the vessel) services connecting multiple European ports.

5.1. Tactical/strategic problems

Maritime transportation involves activities from a wide number of different actors and the number of problems arising in the field is noticeable. From a tactical and strategic point of view, the problems tackled more frequently are the fleet composition problem, the maritime liner problem, the line fleet assignment and the service network design. The fleet composition (and renewal) problem seeks the optimal combination of vessels capable of satisfying the expected demand. The objective of the problem is to identify not only the number and size of vessels, but also which part of the fleet should be chartered and the duration of the chartering contract, and which part should be of property (Pantuso, Fagerholt, & Wallace (2015)). The maritime liner problem is the definition of the line service offered by the company (which ports are visited and with which frequency) and is often connected to the line fleet assignment problem, where vessels are assigned to line services (Wang & Meng (2017a) and Fagerholt (2004)). Service network design (Agarwal & Ergun (2008)) is a particular case of liner service problem and, as described further in this section, it holds great potential for future research. Fleet deployment (Wang & Meng (2012), Alfandari et al. (2019)) and network design (Plum, Pisinger, & Sigurd (2014), Karsten, Pisinger, Ropke, & Brouer (2015)) are all problems for which typically no intermodal integration is considered. A noticeable example of a research combining multiple problems (fleet design, ship scheduling and cargo routing) is Mulder & Dekker (2014), where a composite solution approach is proposed making use of different algorithms (aggregation, disaggregation and a genetic algorithm) to test a real line problem. Comparison with solutions adopted by a great liner shows sizeable improvements. One example of intermodal transportation combining truck services and inland waterways (barges) can be found in Inghels, Dullaert, & Vigo (2016) where it is shown that the use of intermodal transportation using barges can help to reduce both costs and emissions and negative externalities.

Vessel scheduling is an important problem in maritime transportation, but is mainly treated as unimodal. The interested reader can refer to Meng, Wang, Andersson, & Thun (2014) and Dulebenets, Pasha, Abioye, & Kavoosi (2019) which offer a complete overview of the state of the art.

5.1.1. Ship scheduling and cargo routing

In this section we present the formulation proposed by Agarwal & Ergun (2008) for the *simultaneous ship-scheduling and cargo-routing* problem. We chose this formulation as it is general and can be easily adapted to problem variants. The tactical problem of optimizing the ship scheduling consists of creating the service network by designing the ship routes, i.e., defining the sequence of ports to be visited and assigning ships to the sequences. The ships move on cycles from one port to the other and, in normal conditions, follow the same port sequence. The industry standard is to offer weekly departures from each port. Thus, the number of ships required to serve a route is given by the entire length of the route in terms of weeks (e.g. if the total ship turnaround time requires 6 weeks, 6

vessels are needed to maintain the weekly departure target). The design of the service network is called *ship scheduling problem* and each route is called *line*.

In the operational planning stage, a carrier needs to decide if a certain shipping request should be accepted for transportation and using which line, or otherwise reject the request. This problem is known as the *cargo routing* problem.

Decisions taken at the tactical level influence and are influenced by decisions taken at the operational level. The tactical decision of the ship scheduling influences the service availability and capacity of the operational cargo routing problem, and it is at the same time influenced by the cost and revenue information given by the optimization of the operational problem. Agarwal & Ergun (2008), therefore, propose a MIP model that integrates tactical and operational decisions giving a general and compact formulation. The problem is formulated as a multi-commodity flow problem with side constraints on a *space-time* network $G = (N, A)$, where the time horizon is one week. The formulation makes use of the following sets:

- P is the set of ports;
- K is the set of commodities requested. Each commodity request is identified by a triplet $(\tilde{o}, \tilde{d}, \tilde{t})$ where $\tilde{o}$ is the port of origin, $\tilde{d}$ is the port of destination and $\tilde{t}$ is the day of the week in which the demand becomes available;
- V is the set of different types of vessels available in the carrier fleet;
- N is the set of nodes. Each node $i \in N$ represents a port on a day of the week. Thus, for each port in P, we have seven nodes in N. When needed, nodes are referred to as $i_{p,t}$ to specify that the node represents port p at day t;
- A is the set of arcs in the network. It contains different types of arcs:
  - A_g is the set of *ground arcs* connecting a node $i_{p,t}$ with a node $i_{p,t+1}$. These arcs represent vessels and cargo stopping overnight at port p. The set of ground arcs for a particular ship type $v \in V$ is A_g^v;
  - A_s^v is the set of *sea arcs* $(i_{p,t}, j_{q,k})^v$, where $k - t$ is equal to the required transit time by a vessel of type v. If cargo is shipped from port p to port q with different means of transportation (truck, rail, inland waterways etc) the arc is contained in the general set A_s. A_s is the set of all arcs representing a transportation between ports, and is given by $\bigcup_{v \in V} A_s^v$ plus any arc connecting ports not necessarily by maritime transportation;
- $A^-(i)$ is the set of incoming arcs to node i, while $A^+(i)$ is the set of outgoing arcs from node i;
- L^v is the set of feasible service lines. Service lines are feasible when they respect the weekly frequency required by the industry and a number of operational constraints (number of ports visited, ports that can accept vessel type v ...).

The parameters are the following:

- $d^{(\tilde{o},\tilde{d},\tilde{t})}$ is the maximum demand, expressed in TEUs, of commodity $(\tilde{o}, \tilde{d}, \tilde{t})$;
- $\hat{r}^{(\tilde{o},\tilde{d},\tilde{t})}$ is the revenue generated by satisfying one TEU of the demand of commodity $(\tilde{o}, \tilde{d}, \tilde{t})$;
- u^v is the capacity of the vessel type $v \in V$, expressed in TEUs;
- NUM^v is the number of available vessels of type v in the fleet;
- $\tilde{c}_p^v$ is the cost incurred by a ship of type v calling at port p;
- c_p is the cost that a TEU of cargo generates at port p per day;
- c_a^v is the cost generated by a vessel of type v traversing arc a. If $a \in A_s^v$ it represents the cost of the trip, if $a \in A_g^v$ it represents the cost of the overnight stay at port p;
- NV_l is the number of vessels of type v required to serve line $l \in L_v$.

The model uses the following variables:

- $x_l, l \in L^v, v \in V$, is a binary variable which has value equal to 1 if the line service l is activated, 0 otherwise;
- $f_a^{(\tilde{o},\tilde{d},\tilde{t})}, a \in A, (\tilde{o}, \tilde{d}, \tilde{t}) \in K$, is a non-negative, continuous variable representing the flow of commodity $(\tilde{o}, \tilde{d}, \tilde{t})$ on arc a.

The mathematical formulation of the problem is the following:

$$\max \sum_{(\tilde{o},\tilde{d},\tilde{t}) \in K} \sum_{a \in A^-(d)} \hat{r}^{(\tilde{o},\tilde{d},\tilde{t})} f_a^{(\tilde{o},\tilde{d},\tilde{t})} - \sum_{(\tilde{o},\tilde{d},\tilde{t}) \in K} \sum_{(i,j) \in A} c_i f_a^{(\tilde{o},\tilde{d},\tilde{t})} - \sum_{v \in V} \sum_{l \in L^v} \sum_{a \in l \mid p \in a} (\tilde{c}_p^v + c_a^v) x_l \tag{5a}$$

$$\sum_{a \in A^-(i)} f_a^{(\tilde{o},\tilde{d},\tilde{t})} - \sum_{a \in A^+(i)} f_a^{(\tilde{o},\tilde{d},\tilde{t})} = 0 \qquad i \in N, \ (\tilde{o}, \tilde{d}, \tilde{t}) \in K \tag{5b}$$

$$\sum_{(\tilde{o},\tilde{d},\tilde{t}) \in K} f_a^{(\tilde{o},\tilde{d},\tilde{t})} - \sum_{v \in V} \sum_{l \in L^v \mid a \in l} u^v x_l \leq 0 \qquad a \in A_s \tag{5c}$$

$$\sum_{a \in A^-(d)} f_a^{(\tilde{o},\tilde{d},\tilde{t})} \leq d^{(\tilde{o},\tilde{d},\tilde{t})} \qquad (\tilde{o}, \tilde{d}, \tilde{t}) \in K \tag{5d}$$

$$\sum_{l \in L_v} NV_l x_l \leq NUM^v \qquad v \in V \tag{5e}$$

$$x_l \in \{0, 1\} \qquad l \in L_v, v \in V \tag{5f}$$

$$f_a^{(\tilde{o},\tilde{d},\tilde{t})} \geq 0 \qquad a \in A, (\tilde{o}, \tilde{d}, \tilde{t}) \in K \tag{5g}$$

The objective function (5a) maximizes the net profit. Constraints (5b) maintain the flow balance at every node of the space-time network. Constraints (5c) limit the capacity on the line based on the vessel used, while constraints (5d) limit the flow on arcs to the maximum demand. Constraints (5e) ensure that the number of vessels of a certain type used is lower than the maximum number of vessels available for that type. Constraints (5f) and (5g) define the nature of variables.

Various solution methods (heuristics, column generation and Bender's decomposition) are proposed and compared in Agarwal & Ergun (2008). The formulation proposed is the first to jointly model the tactical ship scheduling problem and the operational cargo routing problem and sparked great interest in the literature. For instance, both Wang & Meng (2012) and Song & Dong (2012) take elements from the work proposed by Agarwal & Ergun (2008).

5.1.2. Trends and future research

There is a lack of research on integration of sea transportation with pre-haul and post-haul activities. Recent trends from shipping companies suggest that plans for future vertical integration are beneficial. Most of the largest shipping companies are already main actors in the container terminal industry, like Tanjung Pelepas APM terminal (APM is the same company holding Maersk). MSC bought, in 2015, the Maasvlakte terminal DDN in Rotterdam and COSCO, with the acquisition of OOCL, entered the Long Beach Container Terminal, one of the terminals with the highest traffic in the world. As indicated in Notteboom, Parola, Satta, & Pallis (2017), shipping companies define their line services on the basis of their presence in container terminals, preferring to concentrate activities and transshipment in those ports where they are able to directly super-

vise loading and unloading operations. This shows how impactful could ground operations be in the definition of the line services for companies.

The next step in integration will be to expand it to inland intermodal dry ports. With a foreseeable change of the current modal split in favour of rail transportation, service network design problems should focus on considering the optimization of an integrated network that extends outside the boundaries of maritime transportation to consider dry ports. The location and size of dry ports is a well studied topic. However, integration is missing in other key problems such as network design, service network design, cargo routing, fleet deployment and service scheduling, where considering integration between problems and modalities can potentially offer better solutions when compared to the currently adopted ones. For instance, line services could be designed considering a wider network, implementing intermodal connections between different ports, and cargo can be routed with a connected schedule of maritime and rail services.

One important trend in the unimodal maritime literature is the surge of works trying to integrate multiple problems (port service frequency determination; fleet deployment; sailing speed optimization and vessel schedule design) in order to provide better solutions. Some works aim at implementing tactical and strategic decisions while others focus on linking together multiple tactical problems. Examples can be found in Tan, Wang, Meng, & Liu (2018), Zhen, Wang, Laporte, & Hu (2019) and Pasha et al. (2020). The studies proposed up to this date are focused on maritime specific problems. The integration of multiple problems is relevant and future research should be develpoed in this direction. An even stronger integration involving multimodal aspects (rail service schedule (Yan, Zhu, Lee, Jin, & Wang (2020)), intermodal yard management (Gambardella, Mastrolilli, Rizzoli, & Zaffalon (2001)), VRP for road services) could be explored in order to search for better solutions and further strengthen the multimodal aspect of networks.

5.2. Operational problems

Operational problems focus mainly on finding the optimal cargo route or the set of services needed to minimize the cost for transporting a cargo. Flow formulations are often used as in Agarwal & Ergun (2008) where the authors present an integrated model that solve ship scheduling and cargo routing simultaneously. Balakrishnan & Karsten (2017) propose a MILP formulation for a multi-commodity flow problem on an augmented network with constraints on the maximum number of transshipment operations inspired by real life applications. Zhao, Zhang, & Xue (2016) propose a Binary Linear Programming (BLP) formulation to model a multi-commodity flow problem, seeking the optimization of the container routes in an intermodal network, combined with rail services. Ayar & Yaman (2012) propose various valid inequalities and a Lagrangian relaxation approach for a multi-commodity routing problem with sea and road services. Infante, Paletta, & Vocaturo (2009) study a ship-truck intermodal transportation problem. Optimization of flows is the most studied operational problem in long-haul maritime and waterway transportation. The formulations used are various but can easily be derived from the one presented in Sections 6.2.1 and 5.1.1. One problem worth to be mentioned is the Vessel Recovery problem, where a liner needs to recover the time lost by a vessel due to disruptions. Prime examples of works in the field are Li, Qi, & Lee (2015), Li, Qi, & Song (2016), Cheraghchi et al. (2018) and Abioye, Dulebenets, Pasha, & Kavoosi (2019). Multimodal integration is a new intresting trend, one fine example can be fiund in Abioye, Dulebenets, Kavoosi, Pasha, & Theophilus (2020). problem, however, is mainly treated as unimodal, with little to no integration of multimodal aspects.

5.2.1. Trends and future research

Intermodal operational problems with sea transportation that integrate pre-haul and post-haul operations did not receive much attention in the literature. This is mostly due to the fact that these operations are carried out by completely different actors that rarely collaborate. As already pointed out in Lam & Gu (2013), a promising field for future research is the development of studies focused on an integrated setting where shipping companies cooperate with both ports and hinterland intermodal terminals to offer integrated, synchronized, optimized services. Cooperation between actors vertically placed in the supply chain could bring positive effects to any party involved in intermodal transportation. The use of optimization techniques should be able to prove it. For instance, vessel recovery can potentially improve solutions by re-routing goods in a coordinated intermodal network.

Environmental issues such as congestion, emissions and negative externalities should also be considered since benefits in these aspects are potentially more relevant than those coming from a pure economical analysis. An integrated and coordinated system would be capable of reducing the environmental impact of multimodal transportation with beneficial effects for the society. In general, there is a lack of studies considering multiple objectives, with almost every work focusing on the minimization of transportation cost. However, transshipment and transfer cost are more present than in the past, as witnessed by the comparison to the studies surveyed in SteadieSeifi et al. (2014).

6. Multimodal transportation using more than two modalities

In this section we discuss the works that consider the planning and management of shipments involving more than two modalities. This is very common, for instance, in international maritime transportation, where the shipment is picked up at shipper location using a transportation unit such as a container. Commodities are loaded in the container and then moved to an intermodal railroad terminal. The container is moved through a rail service to the maritime port, where it is loaded on a container vessel and the same pattern is symmetrically repeated at the destination country. The goal of obtaining a better modal shift, less dependent on road transportation, increases the potential interest for problems involving multimodal transportation with three or even more modalities used, and generates the need of properly designed intermodal networks and operational systems. Fig. 5 shows the number of papers studying problems of multimodal transportation with more than two modes, for each combination of modes used, besides the road. Of the studies considered, 8 tackle operational problems, while 23 studies are focused on optimizing tactical and/or strategic issues.

6.1. Tactical/strategic problems

The use of multiple modalities needs reliable and efficient networks in order to be effective. Identifying a proper logistic network is the objective of Logistics Network Design (LND) problems. Ghane-Ezabadi & Vergara (2016) propose the integrated intermodal logistic network design where the objective is the identification of terminal locations and the selection of regular routes and transport modes for commodities. The authors make use of composite variables representing complete paths and find state of the art solutions using a decomposition-based search algorithm and a two-stage optimization algorithm. Zhang, He, Li, & Wang (2017) present a multimodal logistic network design where demand is characterized by time windows for pick up and delivery and environmental concerns in the form of carbon emissions are evaluated. An application of logistic network optimization can be found in Kazemi & Szmerekovsky (2015) where the use of pipelines is considered.

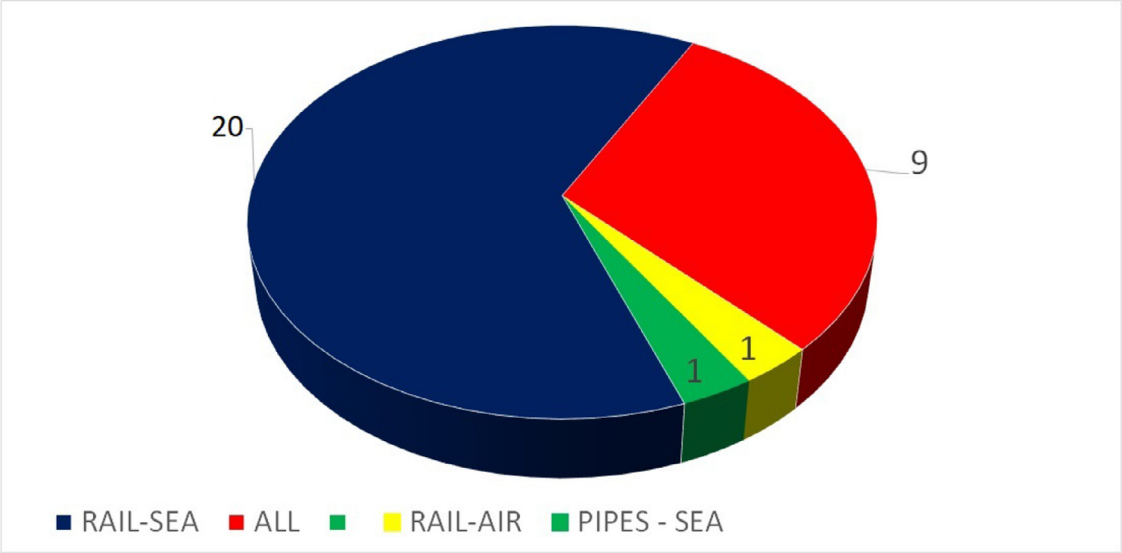


Fig. 5. Modalities combinations.

The authors compare the results of two variants of the model presented, one strictly unimodal while the other is multimodal. The results show that a network properly designed for multimodality allows a reduction of costs and better performances.

The same objective of LND is shared by multimodal network design problems. Dulebenets, Ozguven, Moses, & Ulak (2016) tackle the network design problem for the transportation of perishable products. In their work they propose a MILP formulation and use real world data to test the design of a transportation network that efficiently uses all the available modalities (road, rail, barge, sea and air transportation) in the US and Far East. Non-linear aspects of the formulation are linearized through piecewise linear approximations. Results shows that the rate of decay of perishable product is a strong influence for the proper design of a multimodal network. Zhang, Zhan, Chen, & Li (2018b) show the positive effects of considering the design of a multimodal transportation network, with a focus on emissions and negative externalities. The same authors in Zhang, Eglese, & Li (2018a) study the optimal location for logistics parks in a multimodal regional logistics system using variational inequalities. Network design optimization is studied also in Wang & Meng (2017b), where the problem is relaxed and decomposed in multiple subproblems which are solved using tailor-made algorithms, and Lam & Gu (2016). In the latter, the authors propose a bi-objective optimization model where two conflicting goals (minimization of costs and minimization of transit times) are optimized. Carbon emission and environmental constraints are also considered. The results show that a higher use of rail and inland waterways services can reduce both transportation costs and emissions.

The location component of network design is often limited by the fact that modality exchange locations, like maritime ports or airports, cannot be chosen freely. Henceforth, the study of the optimal design of a service network is far more prominent in the literature. Qu, Bektaş, & Bennell (2016) and Liotta, Stecca, & Kaihara (2015) study intermodal SND problems where the modalities used are rail, road and maritime services (both deep and inland water) with a particular attention to environmental aspects, especially regarding emissions and environmental sustainability. Other contributions to SND applied to intermodal transportation with trucks, trains and barges or inland waterways systems are due to Zhang, Janic, & Tavasszy (2015) and Braekers, Caris, & Janssens (2013). In the former, the authors propose a multi-actor, multi-commodity multimodal network optimization model. The proposed bi-level model optimizes, in the upper level, the government problem of minimizing the total network cost, given by transport cost and emissions, and in the lower-level the shipper/carrier problem of deciding the optimal route in the given network. The solution is found through the use of a genetic algorithm. Demir et al. (2016) study a service network design problem where travel times are uncertain and stochasticity is addressed using a Sample Average Approximation method. The stochastic model is applied to a real case concerning the Danube river and the results show that the solutions found by the stochastic model are robust and perform better than a deterministic model.

Intermodal transportation involves different kinds of problems like identifying the optimal resource allocation (Wang, Wang, & Zhang (2017a)), scheduling (Chen & Schonfeld (2017)), flow optimization (Resat & Turkay (2015)) or transportation planning, especially when a synchromodal approach is considered as in Mes & Iacob (2016) and Resat & Turkay (2019).

Network Design and Service Network Design are the most common problems studied and their formulation are similar to the ones presented in Section 3.1.1 and Section 5.1.1. By defining different characteristics on services and routes the formulations presented can easily be adapted to include multiple modalities.

6.1.1. Trends and future works

In SteadieSeifi et al. (2014) the number of studies tackling tactical and strategic problems making use of more than two modalities was limited to five. Since 2014, there is a clear trend where researchers focused on optimizing logistics networks including multiple modalities and, today, tactical and strategic problems are widely studied and the literature is dense of different formulations and solution techniques. The majority of the studies analyze the problem without making use of real data and relying mostly on simulated instances. Randomly generated realistic scenarios may potentially miss some key characteristics of real data and, therefore, the study of the effects of optimization applied to real world instances would be of interest. Another interesting research direction to follow is the study of more coordinated systems between the different modalities: different problems, like service scheduling and resource allocation (Wang et al. (2017a)), may potentially influence the network design and the flow of goods within the multimodal network. For instance, intermodal terminal yard management can be improved by integrating truck, vessel and train

schedules. At the same time, cargo can be redirected to less congested terminals or take different routes if optimization models take into account the integration among different modalities. Resource management integration is still lacking and, as pointed out in SteadieSeifi et al. (2014), it is a line of research with the potential of enhancing the quality of the solutions found by network design problems. The importance of global networks capable of reacting to disruptive effects was made clear by the Covid-19 pandemic, where personal protection equipment material needed to be distributed efficiently in networks with the key links missing (for instance, air transportation in passenger aircraft shrinked to near zero and the only available options were freighter aircraft and ro-ro services). To the authors knowledge, this area of research is still unexplored when large networks, making use of more than two modalities, are considered. A trend in the design of intermodal networks for the cold supply chain is the consideration of carbon footprint-related objective, as in Ma, Wang, Peng, & Song (2018) and Wang, Tao, & Shi (2018).

6.2. Operational problems

Operational problems in multimodal transportation, with the use of more than two modes, are driven by the need of combining different legs, served by different transportation means, in a cost efficient way, while at the same time maximizing the service offered to customers. The most common problem is the selection of the routes making use of the multiple transportation modes, and focusing on the choice of the different services between those available to move freight from origin to destination throughout a multimodal network.

Similarly to intermodal rail-road transportation, the structure of transportation networks that make use of multiple modes fits perfectly with flow formulations. Thus, the selection of best routes is often modeled as a multimodal multi-commodity flow problem as in Chang (2008), where the MILP formulation proposed is solved through the use of Lagrangian relaxation. Xiong & Wang (2014) propose a bi-level genetic algorithm and Tao, Chew, Lee, & Shi (2017) develop a column generation approach. Darayi, Barker, & Nicholson (2019) propose the design of a network composed by rail, road, sea and air services impacting multiple industries in order to minimize the effects of disruptive events. They propose a multi-commodity network flow formulation and study the optimal re-route after a component of the network is disrupted. A nice theoretical result for multi-commodity network flow problems is shown in Gendron & Gouveia (2016), where the MILP model is reformulated by exploiting discretization. The authors consider the piecewise linear multi-commodity network flow problem, with the addition of a constraint specifying that the total flow on each arc must be integer. Different valid inequalities are proposed. Solutions are identified using Lagrangian relaxation that provides high quality lower and upper bounds. Another interesting solution method using Danzig-Wolfe decomposition in a column generation algorithm is presented in Moradi, Raith, & Ehrgott (2015). In Baykasoğlu & Subulan (2016) the authors propose a MILP formulation for a multi-objective load planning problem with the focus on sustainability, capable of generating solutions tailored for intermodal transportation involving road, rail and sea services. An example of the improvements generated by coordinating services of different modalities can be found in Zhang & Pel (2016), where the authors study a real life scenario, the port of Rotterdam, and make a comparison between the typical intermodal approach and the recently introduced *synchromodal* approach (Tavasszy, Behdani, & Konings (2017)). The problem presented is a scheduling problem where, by exploiting the features of synchromodality, the objective is to coordinate services of different modalities in order to effectively allow a smooth switch between them, reduce dead times, offer a better service to customers and reduce emissions all at the same time. Synchromodality is also addressed in Rivera & Mes (2019), where the authors propose a model to integrate the scheduling of dryage and long-haul (train and barges) transportation. Dryage operations are scheduled using a MILP formulation, while the long-haul schedule is modelled as a Markov Decision Process due to the uncertainty of travel times. Heuristic methods are proposed to integrate the two phases. The results show that integrating end-haul and long-haul schedules can potentially generate improved solutions. Another scheduling problem involving the coordination of trucks, trains and barges transportation services can be found in Behdani, Fan, Wiegmans, & Zuidwijk (2014).

6.2.1. Multi-objective multimodal multi-commodity flow problem

International intermodal third and fourth party logistics operators must face the problem of selecting the best routes for shipments throughout the whole international multimodal network. This problem is often approached using multimodal flow formulations. We now present the formulation proposed in Chang (2008) for a multi-objective, multimodal and multi-commodity flow problem. The formulation is general enough to be adapted to multiple modalities, it considers time windows and a concave representation of costs, making it applicable to real-world situations. The solution technique proposed in the paper is based on decomposing the main problem in smaller, simpler sub-problems, making it a suitable starting point for more specific problems and applications. The problem is to select the best routes for the international shipment of goods. When long distances are considered, air and maritime modalities are predominant and sometimes mandatory choices, while for national or short-medium distances the rail-road intermodal combination is far common. The effect of economies of scale have to be considered. Each arc (i, j) has a cost which is assumed to be a continuous non-convex piecewise linear function of the total flow along the arc (to represent economies of scale). Different modalities have different cost structures, but, generally, they all depend on the total weight transported.

The problem is modelled on a directed graph $G = (N, A)$ where N is the set of nodes, and A is the set of arcs representing connections (transportation services) between different locations. Nodes in set N represent physical locations such as shippers and consignees warehouses, intermodal terminals, ports and airports. Arcs connecting nodes represent transportation services and the modality used depends on the nodes connected (an arc connecting two ports is a maritime service, an arc connecting a warehouse with an intermodal terminal is a road service and so on). Different shipment requests are expressed as different commodities flowing through the graph. Each commodity is, thus, a single shipment.

The formulation makes use of the following sets:

- K is the set of commodities to be transported. Each commodity $k \in K$ is associated with an origin node $\tilde{o}(k)$, a destination node $\tilde{d}(k)$ and a demand d_k;
- $R = \{1, 2, \ldots, |R|\}$ is the set of weight ranges for the different transport modes. Each weight range is associated with a different cost.

The set of parameters is:

- $\bar{c}^r_{ij}$ is the fixed cost associated with arc (i, j) and range $r \in R$;
- c^r_{ij} is the variable cost associated with arc (i, j) and range $r \in R$;
- b^{r-1}_{ij} is the lower bound of range r in arc (i, j);
- b^r_{ij} is the upper bound of range r in arc (i, j).

The objective of the problem is to identify optimal paths for transporting commodities from origins to destinations, to minimize the total transport cost, while satisfying the time window constraints.

The variables used are the following:

- y_{ij}^r is a binary variable equal to 1 if range r is selected on arc (i, j);
- $\bar{f}_{ij}^{kr}$ is a non-negative variable representing the flow of commodity k on arc (i, j), assuming the total flow on the arc is limited within range r;
- f_{ij}^k is a non-negative variable representing the flow of commodity k on arc (i, j).

The formulation is the following:

$$\min \sum_{(i,j)\in A} \sum_{r\in R} \bar{c}_{ij}^r y_{ij}^r + \sum_{(i,j)\in A} \sum_{r\in R} \sum_{k\in K} c_{ij}^r \bar{f}_{ij}^{kr} \tag{6a}$$

$$\sum_{j\in N} f_{ij}^k - \sum_{j\in N} f_{ji}^k = \begin{cases} d_k & if\ i = \tilde{o}(k) \\ -d_k & if\ i = \tilde{d}(k) \\ 0 & otherwise \end{cases} \quad k \in K, i \in N \tag{6b}$$

$$f_{ij}^k \le \sum_{r\in R} \bar{f}_{ij}^{rk} \qquad (i, j) \in A, k \in K \tag{6c}$$

$$\bar{f}_{ij}^{kr} \le d_k y_{ij}^r \quad (i, j) \in A, k \in K, r \in R \tag{6d}$$

$$\sum_{k\in K} \bar{f}_{ij}^{kr} \ge \max\{0, b_{ij}^{r-1} y_{ij}^r\} \qquad (i, j) \in A, r \in R \tag{6e}$$

$$\sum_{k\in K} \bar{f}_{ij}^{kr} \le \max\{0, b_{ij}^r y_{ij}^r\} \qquad (i, j) \in A, r \in R \tag{6f}$$

$$\sum_{r\in R} y_{ij}^r \le 1 \qquad (i, j) \in A \tag{6g}$$

$$y_{ij}^r \in \{0, 1\} \qquad (i, j) \in A, r \in R \tag{6h}$$

$$f_{ij}^k \ge 0 \qquad (i, j) \in A, k \in K \tag{6i}$$

$$\bar{f}_{ij}^{kr} \ge 0 \qquad (i, j) \in A, k \in K, r \in R. \tag{6j}$$

The objective function (6a) minimizes the total cost. Constraints (6b) are the classical flow conservation equations. Constraints (6c) make sure that the flow of commodity k on arc (i, j) is not higher than the total flow of the same commodity in all available ranges on arc (i, j). These constraints are used to link variables f_{ij}^k and $\bar{f}_{ij}^{kr}$. Constraints (6d) force to choose range r on arc (i, j) when there is a positive flow of at least one commodity through it. Constraints (6e) and (6f) set the lower and upper limits for the total flow on arc (i, j) if range r is chosen and force variable $\bar{f}_{ij}^{kr}$ to 0 if range r is not chosen. At most one range is selected for each arc due to constraints (6g). Constraints (6h)–(6j) define variable domains.

The most common technique proposed in the literature to solve the piecewise linear and concave multi-commodity network flow problem is Lagrangian relaxation. Chang (2008) exploits this technique and derives a Lagrangian problem by dualizing constraints (6c) using non-negative multipliers α_{ij}^k for all commodities k and arcs (i, j).

The formulation proposed by Chang (2008) generalizes the problem and introduces features and properties further developed by multiple authors like Baykasoğlu & Subulan (2016) and Zhang et al. (2017).

6.2.2. Trends and future research

Operational problems are complex and finding good solutions is a difficult task. The use of operational research techniques can be a powerful tool in the hands of operators since it can help them find better solutions quickly. Despite that, operational problems are not well studied, with the literature focused mainly on tactical and strategic problems. Researchers interested in the topic should explore ways to solve the problems that consider coordinated and combined systems of different modalities. Moreover, integration can be expanded by considering problems related to warehouse management, yard management for intermodal terminals, service scheduling for handling equipment and other logistics problems not strictly related to transportation. In this sense, the growth of the research in synchromodal transportation is a promising trend that should continue in the future. In a world where data driven decisions are more and more prominent, both present and future technologies allow researchers to undertake the challenge of building and optimizing fully synchronized systems. Giusti, Manerba, Bruno, & Tadei (2019) propose a definition of synchromodal transportation and examine the key factors for its success. Dynamic optimization of commodity flows throughout a global network, considering multiple modalities involved and where actors share information and collaborate to reach global optimization, is the goal to aim to in the future. A recent contribution addressing this aspect is Qu, Rezaei, Maknoon, & Tavasszy (2019) where the authors tackle the problem of rescheduling traffic flows in a synchromodal setting, with tests based on data gathered from the port of Rotterdam.

In general, the number of studies considering multi-commodity and multi-objective aspects, mainly in the form of carbon emission minimization, increased if compared to the state of the art pictured by SteadieSeifi et al. (2014).

7. Conclusions

Multimodal transportation is an increasingly important topic, of interest to policy makers, transportation operators and shippers. Unfortunately, unimodal transportation is still prominent. Environmental aspects are more and more considered in policy and research (the number of studies considering carbon emission objectives and constraints increased in the last years) and this trend should be fostered, as awareness on the importance of environmental features in optimization of logistics and transportation is fundamental. Capturing the trade-offs involving environmental aspects is a key factor that will make research applicable to real-world instances. For instance, the maritime company CMA-CGM is heavily investing on the renewal of its fleet and predicts to be able to operate 20 new LNG-powered vessels, 9 of them with capacity of 22,000 TEUs. This is a response to the recently active regulation IMO 2020, which has the objective to substantially reduce the amount of sulphur oxide emanated by vessels. The IMO 2020 regulation is an example of a recent trend where big companies are required to sacrifice profits in order to foster sustainability. Prior to this regulation, big companies were encouraged to present their sustainability plan. However, this was done through voluntary disclosure to satisfy stakeholder and market expectations. The majority of companies still give priority to profit, with little consideration of environmental aspects. While environmental awareness is spread in both the scientific literature and the public opinion, governments and international organizations have only recently started to take serious actions to support envinronmental sustainability.

The analysis of the research made in this literature review led to the following main conclusions:

- The number of studies considering and implementing transshipment cost increased since the literature review in SteadieSeifi et al. (2014). This is an important trend that needs to continue. A precise estimation of transshipment cost is fundamental in obtaining correct solutions and, as shown in Hanssen et al. (2012), is one of the factors influencing the set-

ting for which multimodality is more convenient than unimodal options. On the other side, not enough effort was put in considering congestion at terminals (to the best of the authors knowledge, only Assadipour et al. (2015) directly tackle the issue in a very specific problem where hazardous materials are involved). Contemplating congestion at yard is important when the route of cargo is to be planned in multimodal networks.

- Network design, service network design, flow and transportation problems are common to all modality combinations, with the exception of problems with drones, where routing is instead the underlying problem. This generates similarities in the solution approaches proposed for the different problems, even with different modalities involved. For instance, in network/service network design problems, a common practice is the use of decomposition methods. Application of these techniques has been observed in problems involving different modalities (Ghane-Ezabadi & Vergara (2016), Chang (2008), Agarwal & Ergun (2008), Ayar & Yaman (2012), Wang & Meng (2017b), Poudel et al. (2016)). Decomposition techniques can successfully be applied to transportation problems too, as shown in Moradi et al. (2015). While some problems, like vessel schedule recovery, are very modality specific, the concept of synchromodality suggests a holistic approach, as shown in Qu et al. (2019) where the resolution of shipment flow rerouting leads to a transshipment organization in the intermediate terminals and to the corresponding service rescheduling. Optimizing at the same time multiple problems on multiple modalities can lead to a transversal use of optimization techniques.
- Optimization in multimodality between rail and road is a commonly studied topic and a fertile field for applied research. Since the aim of a large part of developed countries is to change the current distribution of the modal split, more and more research on exploiting the benefits of multimodality can be applied to real world scenarios. Unfortunately, the majority of the studies do not consider the fact that multiple actors are involved in multimodal transportation and that their objective could not coincide. With the exception of Chen & Schonfeld (2017), no other studies explore the effects of collaboration and competition behaviours in multimodal transportation. This is an interesting field of research since it proves that, by cooperating, multiple actors can effectively improve the network and the logistics systems currently in place.
- Intermodal studies combining sea transportation with other modalities, on the other side, deserve more attention. Trends in maritime transportation seem to indicate that in the future there will be an increased vertical integration between the sea and land side of multimodal transportation. In fact, optimization by maritime companies is affected by their financial presence at container terminals. A combined, connected and integrated sea-land network, where transportation services are synchronized in order to reduce waiting time for vehicles, idle times at terminal and negative externalities such as pollution, congestion and community severance, is the goal to be achieved in the coming years.
- From a methodological point of view, the solution of network design and service network design problems mostly relied on tabu search heuristics. Recently, we witnessed a surge of metaheuristics, such as Genetic Algorithm and GRASP. This is an interesting trend, as exploring multiple classes of metaheuristics may lead to the use of new and potentially superior techniques capable of improving the quality of solutions. In particular, metaheuristics seems capable of finding satisfying solutions in contained times even for great instances. This is an important feature needed to implement optimization techniques to real world applications. SteadieSeifi et al. (2014) pointed out a lack of approximation techniques in solving these problems. The trend mentioned above shows that the research is moving in this direction.
- Due to its recent spike of interest, an increase of studies focused on optimization of multimodal air transportation is expected. E-commerce and the increasing need to move goods from one part of the world to another with limited transit times raised the importance of transportation by means of air mode. Currently, the most studied problem is the optimization of the use of combined truck and drone transportation in order to carry out last mile deliveries. The field of optimization in drone deliveries is young and holds much potential, especially considering that real-world applications were mostly ignored, with the only exception of Jeong et al. (2019).
- The number of studies on multi-commodity problems increased when compared to SteadieSeifi et al. (2014). Commodities have different characteristics and different needs. Thus, considering multiple aspects may lead to better solutions, especially in the assignment of commodity flows throughout a network. Similarly, the number of multi-objective studies increased, with multiple efforts to model the trade-off between delivery time and transportation costs, with the additional objective of minimizing carbon emissions. Darayi et al. (2019) study multi-industry effects of transport planning in their work, considering multiple objectives to be balanced. Dealing with multiple objectives is challenging, both in terms of modeling and in terms of techniques needed to find optimal solutions, and additional research is needed. Also, different objectives should be considered, including for instance other negative externalities or conflicting objectives of the multiple actors involved in multimodal transportation.
- Synchromodal transportation saw a rise of interest and the reasons are easy to understand. The goal of synchromodal optimization is the definition of a synchronized, seamless transportation system where operators share information in a collaborative way and technologies allow the switch of modalities based on the current status of the network, dynamically updated on the go. While this idealistic vision is far from the present situation of transportation with multiple modes, the goal is to obtain an optimized and fully synchronized system.
  In order to achieve fully integrated transportation services using different modalities a great deal of changes are due in both organizational processes and technologies. While on one side this is an obstacle to the creation of integrated systems, on the other side it is an opportunity for the development of new organizational patterns and technologies implementing optimization techniques.
- In general, applications of practical scenarios are increasing as more than half of the studies reviewed are based on real life applications. On one hand the risk is to have models and solution techniques overfitting the specific problem tackled without any degree of generalization, but the goal of proposing formulations and solution techniques capable of tackling real world instances needs to be pursued. A general lack of benchmark instances is noticeable while understandable, since many problems are too specific to allow the generation of benchmark instances.

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